32 Building custom MVC and Razor Pages components

This chapter covers

- Creating custom Razor Tag Helpers
- Using view components to create complex Razor views
- Creating a custom DataAnnotations validation attribute
- Replacing the DataAnnotations validation framework with an alternative

In the previous chapter you learned how to customize and extend some of the core systems in ASP.NET Core: configuration, dependency injection (DI), and your middleware pipeline. These components form the basis of all ASP.NET Core apps. In this chapter we're focusing on Razor Pages and Model-View-Controller (MVC)/API controllers. You'll learn how to build custom components that work with Razor views. You'll also learn how to build components that work with the validation framework used by both Razor Pages and API controllers.

We'll start by looking at Tag Helpers. In section 32.1 I show how to create two Tag Helpers: one that generates HTML to describe the current machine and one that lets you write `if` statements in Razor templates without having to use C#. These will give you the details you need to create your own custom Tag Helpers in your own apps if the need arises.

In section 32.2 you'll learn about a new Razor concept: view components. View components are a bit like partial views, but they can contain business logic and database access. For example, on an e-commerce site you might